Evaluation-Driven Descent, Encoding Permanence, and the Structural Invariants of Intelligence

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Abstract

We establish four linked results for any entity with structure (K, A, σ, γ) persisting under bounded active context, survival pressure, and positive novelty rate $\nu(E) > 0$. First, agency ($\gamma > 0$) necessarily produces evaluation-driven state revision—monotone local descent on expected uncertainty $H(W \mid K)$—where each informative update reduces $H(W \mid K)$ by at most the information gain $G(a, K)$, with equality under exact Bayesian conditioning. Under non-stationary task distribution, descent is local and instantaneous rather than globally convergent. Second, the certainty threshold for encoding an action at level L_i is $\theta_i = 1 - \varepsilon/C_i$, which derives the learning rate hierarchy from cost-of-failure differences rather than assumption. Third, separating the action space A from the higher-order function space M is selectively favored under bounded context: collapsing them is possible but increasingly costly under positive novelty rate, so competition eliminates collapsed architectures. Fourth, $\gamma > 0$ and learning efficiency $\eta_M > 0$ are the unique structural invariants of persistence: the free-context exhaustion argument shows that $\gamma = 0$ entities are eliminated under positive novelty rate; without $\eta_M > 0$, active context fills without relief and inference degrades to failure. Every other property can equal zero for some persistent entity. Therefore $I(E) = \gamma \cdot \eta_M$—the rate of adaptation to novelty—is the unique measure of intelligence, derived not stipulated.

1 Introduction

This paper completes a chain of results deriving the necessary architecture of intelligence from a single generative constraint: bounded active context.

Paper 1 [17] established that any entity accumulating propositions under bounded context necessarily maintains a directed graph with typed edges—where "necessary" means selected for under competition, not logically impossible to avoid (see [17], Remark 1). Paper 2 [18] established that bounded context under growing action count requires convergence toward escalation. Paper 3 [19] established that bounded context forces the normalization of shared propositions from action into a separated K, with an asymmetric retention criterion.

Three questions remain. First: what dynamics govern the updating process when an entity with agency acts on the world and receives feedback? Second: what determines the threshold at which a proven action earns promotion to a more permanent encoding level? Third: what properties must every persisting entity have?

We answer all three. Theorem 1 establishes that agency ($\gamma > 0$) with feedback produces evaluation-driven state revision—monotone local descent on expected uncertainty $H(W \mid K)$. Definition 12 establishes that the certainty threshold for encoding at level L_i is $\theta_i = 1 - \varepsilon/C_i$, deriving the learning rate hierarchy from cost-of-failure differences. Theorem 3 establishes that the action space A and the higher-order function space M are type-irreducible. Theorem 5 establishes that $\gamma > 0$

and learning efficiency $\eta_M > 0$ are the unique structural invariants of persistence, and Corollary 6 derives $I(E) = \gamma \cdot \eta_M$ as the unique intelligence measure.

Several lines of prior work converge on these questions. Dupoux, LeCun, and Malik [6] propose a bilevel optimization framework but treat the learning rate relationship between levels as a design parameter rather than deriving it, and deliberately avoid defining intelligence. Hutter [13] and Legg and Hutter [16] define intelligence as average performance over computable environments; we define it as rate of improvement, independent of current performance level. Chollet [4] measures skill-acquisition efficiency, structurally close to our η_M but treated as a benchmark metric rather than a structural invariant. Schmidhuber [22] identifies curiosity as compression progress but treats it as a design objective rather than a necessary condition of persistence. Friston's free energy principle [9] derives minimization behavior from self-organization but does not distinguish $\gamma = 0$ from $\gamma > 0$ within its framework. Cover and Thomas [5] provide the information-theoretic foundations that anchor our convergence argument. The contribution of this paper is to derive, via selection and exhaustion arguments, the unique structural invariants of persistence and the intelligence measure they determine.

2 Formal Framework

We introduce the definitions required for the four main results, extending the framework of [17].

Definition 1 (Entity, World, Uncertainty)**.** An entity $E = (K, A, \sigma, \gamma)$ consists of a knowledge base K (propositions about world W), an action space A (executable actions), a selection mechanism σ that chooses actions given context (which presupposes sensing capacity; see Theorem 5), and an agency parameter γ. The world state space W is a measurable space; the true state $w \in W$ is not directly observable. The *uncertainty* of E about state w given knowledge K is:

$$U(w, K) = -\log P(w \mid K)$$

The expected uncertainty is the conditional entropy:

$$H(W \mid K) = \mathbb{E}_w[U(w, K)] = -\sum_w P(w \mid K) \log P(w \mid K)$$

Definition 2 (Agency)**.** Agency $\gamma \in [0, 1]$ is the probability that the entity engages its adaptive capacity M in response to a learning signal. Equivalently, γ is the response gate: the fraction of environmental signals that trigger an update to (A, K).

The operational signature of $\gamma > 0$ is that the entity bounds uncertainty beyond what immediate survival requires, freeing context capacity for novel states. An entity with $\gamma = 0$ responds to no signal and accumulates no free context. An entity with $\gamma > 0$ engages M on signals, resolves uncertainty, promotes actions to lower encoding levels, and thereby maintains $C_{\text{free}} > 0$ (Definition 7).

The distinction between $\gamma = 0$ and $\gamma > 0$ is not between entities that face threats and those that do not. It is between entities that build reserve capacity for states they have not yet encountered and those that do not.

Definition 3 (Action Feedback)**.** For action a executed in world state w, the *feedback* $\varphi(a, w)$ is the observable consequence. The entity updates its knowledge base: $K_a = K \cup \{p_a\}$, where p_a is the proposition encoding the feedback. The *prediction-error signal* is:

$$\delta = \varphi(a, w) - \mathbb{E}[\varphi(a, \cdot) \mid K]$$

Definition 4 (Information Gain)**.** When entity E executes action a in world state w, it receives feedback $\varphi(a, w)$ encoded as proposition p_a (Definition 3). The *information gain* is the mutual information between the feedback proposition and the world state, conditioned on current knowledge:

$$G(a, K) = I(p_a; W \mid K) \geq 0$$

Non-negativity follows from the definition of mutual information [5]. $G = 0$ if and only if p_a provides no information about W beyond what K already contains.

Definition 5 (Novelty Rate)**.** The *novelty rate* $\nu(E)$ of entity E in world W is the rate at which E encounters world states w where $U(w, K) > 0$:

$$\nu(E) = \lim_{T \to \infty} \frac{1}{T} \big| \{ t \leq T : U(w_t, K(t)) > 0 \} \big|$$

Novelty rate is a property of the gap between W and K, not of W alone. A world with high entropy but an entity whose K covers all reachable states has $\nu = 0$. The distinction matters: entropy is a property of the environment; novelty is a property of the entity–environment coupling. The elimination argument (Theorem 5) requires that novelty include *survival-relevant* states: states where $U(w, K)$ can exceed U_{lethal} unless K is updated. Under positive entropy in W, the world state distribution drifts, and previously non-lethal gaps in K eventually become lethal. The assumption is that $\nu > 0$ includes eventual survival-relevant novelty, not merely benign uncertainty.

Definition 6 (Encoding Hierarchy)**.** The encoding space $\mathcal{E}(c, r)$ is a continuous space parameterized by cost c and reversibility r. Within this space, interfaces $L_0, L_1, \ldots, L_n$ define discrete encoding levels with increasing cost and reversibility:

- L_0 (genomic): highest cost of error, lowest reversibility, slowest update rate.
- L_n (active context): lowest cost of error, highest reversibility, fastest update rate.

Escalation [18] pushes proven actions upward toward more permanent levels. The present paper derives the threshold that governs this promotion.

Definition 7 (Free Context)**.** The *free context* of entity E at time t is the active context capacity not consumed by unresolved content:

$$C_{\text{free}}(t) = C_n - |\text{ctx}_{\text{unresolved}}(t)|$$

where $\text{ctx}_{\text{unresolved}}(t)$ is the set of propositions in active context that have not yet been encoded at a lower level or resolved by M. Free context is the entity's reserve capacity for handling novel states. The architecture derived in Papers 1–3—graph structure, escalation, retention criterion—exists to maximize C_{free}: each result describes a mechanism that moves resolved content out of L_n, preserving capacity for what the entity has not yet encountered.

Definition 8 (Higher-Order Function Space M)**.** Let $A : K \times W_{\text{obs}} \to \Delta\mathcal{A}$ be the action space of informed actions, where W_{obs} denotes observable world states and $\Delta\mathcal{A}$ the simplex over available actions. The *higher-order function space* is

$$M : (A, K) \times L \;\to\; (A, K)$$

where L is a learning signal (feedback from W following action). M is categorically different from A: elements of A map knowledge and observations to action distributions; elements of M map action–knowledge pairs and learning signals to updated action–knowledge pairs.

M contains whatever adaptive capacity has not yet been committed to specific knowledge or action. As the entity learns, components of M resolve into concrete elements of K or A—the same normalization by which shared propositions precipitate from action into knowledge [19]. What remains in M is the entity's capacity to produce further such resolutions.

The hierarchy is open upward: $M^{(2)}$ operates on $M^{(1)}$, yielding meta-learning; $M^{(n+1)}$ operates on $M^{(n)}$. No natural termination point is established (see Section 10).

Definition 9 (M Efficiency)**.** Given a learning event where $(A', K') = M((A, K), L)$, define the *learning efficiency*:

$$\eta_M \;=\; \mathbb{E}_{L|\text{engaged}}\left[\frac{\max_{a' \in A'} G(a', K') - \max_{a \in A} G(a, K)}{|L|}\right]$$

The expectation is taken over learning signals L *conditional on the entity engaging* M (i.e., conditional on the gate γ opening). This conditioning is essential: γ may be directed, so $\mathbb{E}[\cdot \mid \text{engaged}]$ may differ from the unconditional expectation. Conditioning on engagement ensures that the product $\gamma \cdot \eta_M$ is the expected improvement rate by the law of total expectation, regardless of whether γ is content-blind or directed.

Both numerator and denominator are measured in bits; η_M is dimensionless. The three regimes:

- $\eta_M > 0$: the entity learns—grounding improves per unit of learning signal, conditional on engagement.
- $\eta_M = 0$: engagement produces no improvement.
- $\eta_M < 0$ temporarily: exploration, creative destruction, locally irrational steps that may increase long-run survival. Persistently $\eta_M < 0$: the entity does not persist.

Remark 1 (Measurement). Computing η_M requires evaluating $G(a, K) = I(p_a; W \mid K)$, which depends on the true joint distribution $P(p_a, W, K)$. The entity cannot compute its own η_M; it would need the very information it is trying to learn. Both γ and η_M are measurable by an external observer with access to the entity's trajectory and outcomes.

Definition 10 (Intelligence)**.** For an entity E with agency γ and learning efficiency η_M:

$$I(E) = \gamma \cdot \eta_M$$

Intelligence is the rate of adaptation to novelty: the expected rate at which the entity reduces uncertainty about novel states. This is stated here as a definition. Theorem 5 establishes that $\gamma > 0$ and $\eta_M > 0$ are the unique structural invariants of persistence under positive novelty rate.

Definition 11 (Persistence)**.** An entity E *persists* if it survives for all t: there is no finite T at which $U(w_T, K(T)) > U^d_{\text{lethal}}$ terminates the entity. Persistence is indefinite survival, not survival to some fixed horizon.

3 Monotone Descent on Uncertainty

Theorem 1 (Evaluation-Driven State Revision)**.** *Let $E = (K, A, \sigma, \gamma)$ be an entity with $\gamma > 0$ persisting in world W under bounded active context C_n.*

Part I. *Every executed action generates a prediction-error signal, and $\gamma > 0$ guarantees engagement with that signal, producing evaluation-driven state revision—monotone local descent on expected uncertainty $H(W \mid K)$ within the entity's model. Under competition (Remark 1 of [17]), entities that fail to descend are eliminated: descent is selectively necessary, not merely optimal.*

Part II. *Under positive novelty rate $\nu(E) > 0$ (Definition 5), descent never terminates globally: the entity continues to encounter states where $U(w, K) > 0$, so actions with positive information gain always exist.*

Proof. **Part I: Forced Signal, Response Gate, and Convergence.**

Every executed action a generates feedback $\varphi(a, w)$ from the world. The prediction error $\delta = \varphi(a, w) - \mathbb{E}[\varphi(a, \cdot) \mid K]$ is produced at every action regardless of γ.

Agency γ is the response gate. An entity with $\gamma = 0$ receives every signal δ but never updates: (A, K) remain static. An entity with $\gamma > 0$ engages M with probability γ on each signal, integrating the feedback into K by forming $K' = K \cup \{p_a\}$. K is a set of propositions, not a parameter vector, so $\nabla_K U$ is undefined: there is no differentiable loss surface over K. The descent property is instead established directly via the Lyapunov function $H(W \mid K)$.

Step 1: Each update reduces the entity's estimate of $H(W \mid K)$. When the entity encounters a novel state and executes action a, the feedback p_a is informative: $G(a, K) = I(p_a; W \mid K) > 0$. When M integrates this feedback, forming $K' = K \cup \{p_a\}$:

$$H(W \mid K') = H(W \mid K) - G(a, K)$$

This identity follows from the chain rule for mutual information under exact Bayesian conditioning: $H(W \mid K') = H(W \mid K, p_a)$.

The distinction between model and reality. The descent is on the entity's *model* of W, not on the true state of W. Adding a proposition to K refines the posterior within the entity's model—this is monotone by construction, since adding a constraint to the joint likelihood L_K cannot broaden the posterior (see the model-vs-reality remark in [19], Lemma 1). But the proposition may be wrong: the evidence may be noise, the likelihood function may be mis-specified, or the world may have shifted since the evidence was collected. In such cases, the entity's model diverges from reality even as its *estimated* H decreases.

This is not a gap in the argument but the precise content of the persistence conditions established in Theorem 5. The descent on the entity's model is guaranteed at every step. Whether that descent tracks reality depends on the quality of evidence integration—which is η_M. An entity with $\mathbb{E}_t[\eta_M] > 0$ improves its model's correspondence with W on average over its trajectory. An entity whose model systematically diverges ($\mathbb{E}_t[\eta_M] < 0$ persistently) is exactly the failure case that Theorem 5, Part 2 eliminates. Temporary divergence ($\eta_M < 0$ at individual steps) is compatible with persistence and is the mechanism of exploration (Remark 3).

The Lyapunov argument therefore has two levels: (i) within the model, descent is monotone at every informative step; (ii) across the model-reality interface, descent tracks reality only in expectation, guaranteed by $\mathbb{E}_t[\eta_M] > 0$. Level (i) is proved here. Level (ii) is the persistence condition.

Step 2: Monotone convergence within the model. Consider the sequence $\{H_t\} = \{H(W \mid K(t))\}$ indexed by informative update steps, where H is the entropy *as estimated by the entity's model.* By Step 1, $\{H_t\}$ is strictly decreasing at each informative step. By definition, $H(W \mid K) \geq 0$, so the sequence is bounded below. By the monotone sequence theorem, $\{H_t\}$ converges. Since the

sequence is non-increasing and convergent:

$$\sum_{t=0}^{\infty} G(a_t, K(t)) \leq H(W \mid K(0)) < \infty$$

Therefore $G(a_t, K(t)) \to 0$ as $t \to \infty$. The entity converges to a state where available actions yield diminishing informational returns *within its current model*.

Part II: Non-Termination Under Positive Novelty.

When $\nu(E) > 0$, the entity encounters states w with $U(w, K(t)) > 0$ at positive rate. Each such encounter produces an action a with $G(a, K(t)) > 0$. Therefore at every finite t there exists some a' with $G(a', K(t)) > 0$. Descent converges locally (Part I) but never terminates globally.

Non-termination does not require unbounded world complexity. A finite world with $\nu(E) > 0$ suffices: even if W is finite, the combinatorial space of entity-environment interactions exceeds what K can index under bounded C_n.

Interaction with the curation regime. The convergence argument in Part I assumes K grows monotonically. In the curation regime established in [19], $|K|$ is bounded and every addition requires displacement. The Lyapunov function must be refined. Raw $H(W \mid K)$ is the wrong quantity: displacing a proposition about an irrelevant domain in favor of one about a survival-critical domain could increase $H(W \mid K)$ overall while improving the entity's survival prospects. The correct Lyapunov function is *survival-weighted expected uncertainty*:

$$H_{\mathcal{T}}(W \mid K) = \mathbb{E}_{w \sim \mathcal{T}}[U(w, K)]$$

where $\mathcal{T}$ is the entity's task distribution over survival-relevant world states.

Under the cost-benefit displacement criterion of [19], proposition p is displaced by q only when

$$\mathbb{E}[B(q, \mathcal{T})] - c_{\text{retain}}(q, K) \;>\; \mathbb{E}[B(p, \mathcal{T})] - c_{\text{retain}}(p, K),$$

so each displacement reduces $H_{\mathcal{T}}$.

$\mathcal{T}$ itself is non-stationary: as the entity learns and the environment shifts, the distribution of survival-relevant states changes. Under non-stationary $\mathcal{T}$, the global Lyapunov convergence argument does not apply: $H_{\mathcal{T}_t}$ at step t and $H_{\mathcal{T}_{t+1}}$ at step $t+1$ evaluate different functions when $\mathcal{T}$ drifts. What the curation regime establishes is *local, instantaneous* descent: each displacement reduces $H_{\mathcal{T}}$ evaluated against the current $\mathcal{T}$ at the moment of displacement. This is a weaker claim than global convergence, but it is the operationally relevant one: the entity cannot plan against future task distributions it has not yet encountered. The non-stationarity of $\mathcal{T}$ is precisely what drives non-termination (Part II): the target keeps moving, so the entity never reaches $H_{\mathcal{T}} = 0$ globally.

Remark. An entity with $\gamma = 0$ never updates. As W changes, its fixed K becomes increasingly inaccurate: $U(w_t, K)$ grows until it exceeds U_{lethal}. Selection removes such entities. Therefore $\gamma > 0$ is necessary for persistence, a claim established fully in Theorem 5. □

4 Encoding Permanence

Definition 12 (Encoding Permanence Scales with Rigor)**.** Let $L_0, \ldots, L_n$ be encoding levels with associated failure costs $C_0 > C_1 > \cdots > C_n$ and reversibility costs $r_0 > r_1 > \cdots > r_n$. Let ε be the maximum acceptable expected error cost. Then the minimum certainty threshold for encoding an action at level L_i is:

$$\theta_i = 1 - \frac{\varepsilon}{C_i}$$

Derivation. If θ is the certainty that action a is correct, the expected error cost is $(1-\theta)\cdot C_i$. Encoding is safe when:

$$(1-\theta)\cdot C_i \leq \varepsilon \quad \Longrightarrow \quad \theta \geq 1 - \frac{\varepsilon}{C_i}$$

The minimum safe threshold is $\theta_i = 1-\varepsilon/C_i$. This is arithmetic, not a deep result. The substantive content is the *identification*: that the certainty threshold is determined by cost-of-failure, not by a free parameter. Given that identification, the formula follows in one line.

Consequence. At L_0 (genomic), C_0 is the cost of lineage extinction, so $\theta_0 \approx 1$: near-certainty is required. At L_n (active context), C_n is small, so θ_n is meaningfully less than 1: the expected benefit of trying exceeds the expected cost of failure at moderate confidence.

Error propagation through the hierarchy. Under escalation [18], errors partition into *handled errors*, resolved at level L_i, and *unhandled exceptions*, which escalate. The error rate seen by L_{i+1} is the rate of genuinely irreducible errors, not a compounded Bernoulli product of lower-level rates. The thresholds θ_i are independent by construction.

4.1 Deriving the Learning Rate Hierarchy

The learning rate at level L_i is inversely related to the rigor threshold. Each trial of action a yields a binary outcome (success or failure), so the empirical success rate $\hat{p}$ after n trials is the sample mean of n independent Bernoulli random variables in $[0,1]$. Hoeffding's inequality [12] applies directly to sample means of bounded random variables:

$$P(|\hat{p}-p| \geq \delta) \leq 2\exp(-2n\delta^2)$$

The entity requires confidence θ_i that $\hat{p}$ is within δ of p. Setting the failure probability $2\exp(-2n\delta^2) \leq 1-\theta_i = \varepsilon/C_i$:

$$n_i \;\geq\; \frac{\log(2C_i/\varepsilon)}{2\delta^2}$$

The effective learning rate is inversely proportional to n_i:

$$\eta_i \;\propto\; \frac{1}{n_i} \;\propto\; \frac{\delta^2}{\log(C_i/\varepsilon)}$$

Remark 2 (Level-independence of δ). The deviation tolerance δ measures how close the empirical success rate must be to the true rate—a property of measurement accuracy, not of the stakes. Assuming δ constant across levels is the conservative choice: a rational entity might demand tighter precision at L_0 ($\delta_0 < \delta_n$), which would make η_{evo} even smaller relative to η_{ind}, strictly strengthening the hierarchy. Level-independent δ produces the *weakest* version of the learning rate separation.

Since $\theta_0 \gg \theta_n$, it follows that $\eta_{\text{evo}} \ll \eta_{\text{ind}}$. The learning rate difference between evolution and individual learning is *derived* from the cost-of-failure structure, not assumed as a design parameter.

Definition 13 (Certainty)**.** Action a is *certain* at level L_i when $P(a \text{ is adaptive} \mid K) \geq \theta_i$ across sufficient variation in w. Once θ_i is met, the action is encoded at L_i and no longer consumes L_n resources, freeing active context for the next epistemic frontier.

Corollary 2 (Bounded Adaptation)**.** *Encoding at L_i is durable, not literally irreversible: environmental shift can render it maladaptive, but correction requires effort proportional to r_i. Individual learning at L_n discovers actions via rapid evaluation-driven descent (η_{ind} large). Encoding permanence determines which promote to L_0: only those meeting $\theta_0 \approx 1$. This is the Baldwin effect [11], here derived rather than observed.*

5 Convergence to A/M Stratification

Theorem 3 (Convergence to A/M Stratification)**.** *Systems that collapse the action space A and the learning space M into a single parameter space pay an increasing cost under bounded context. Under bounded C_n with positive novelty rate, stratified A/M separation is selectively favored: collapsed systems pay higher costs and are eliminated under competition ([17], Remark 1).*

Proof. The type signatures of A and M are distinct:

$$A : K \times W_{\text{obs}} \to \Delta\mathcal{A}$$
$$M : (A, K) \times L \to (A, K)$$

A maps knowledge and observations to action distributions. M maps action–knowledge pairs and learning signals to updated action–knowledge pairs. These are different functions serving different roles. The question is not whether a system *must* separate them—it need not—but what happens when it does not.

The degenerate case. A system with no separation at all has a single function type $T : W_{\text{obs}} \to \Delta\mathcal{A}$. This is the case $K = \varnothing$: no propositions are separated from the action policies, so every action node carries its own copy of whatever world-model information it needs. Shared propositions are duplicated across nodes rather than extracted—the unfactored state of [17]. The system functions but its nodes are heavy: as the action repertoire grows, the redundant weight grows with it.

Factoring K out of A converts that redundant node weight into edges—each action links to shared propositions rather than storing its own copy. Separating M does the same for learning: update logic is extracted once rather than embedded in every action. This is exactly the normalization argument of [19] applied to function types rather than propositions.

More generally, M, A, and K can all be unified into a single parameter space. The question is efficiency under bounded context, not logical possibility.

The cost of collapsing A and M. A system that uses the same parameters for both action selection and learning faces a structural conflict: updating parameters to incorporate new learning (the M function) modifies the same substrate that encodes stable action policies (the A function). Without stratification, every learning update is an unscoped modification to the action space. This is precisely the mechanism identified in [17]: without stored adjacency between what is being updated and what depends on it, updates cascade uncontrollably.

The empirical manifestation is catastrophic forgetting [14]: neural networks that learn new tasks degrade performance on previously learned tasks because the weight updates that encode new knowledge overwrite the weights that encode old action policies. The network cannot scope the update because it lacks the structural separation between "what I know how to do" (A) and "how I change what I know" (M).

Stratification under bounded context. Under bounded C_n, the cost of catastrophic forgetting is not merely wasteful but fatal. Each unscoped update that degrades a proven action forces that action back to L_n for re-learning, consuming the context capacity that the entity needs for novel problems. Under positive novelty rate $\nu > 0$, novel states arrive continuously. An entity that must re-learn previously encoded actions while simultaneously handling novel states exhausts C_{free}.

Stratification resolves this: A encodes stable action policies at lower encoding levels (protected from learning updates), while M operates at L_n on the current learning frontier. Updates to M do not modify A; modifications to A occur only through the promotion mechanism of Definition 12, gated

by the certainty threshold θ_i. This is the same pattern as the series: not a logical impossibility of collapsing A and M, but convergence under bounded-context pressure toward separation because the collapsed alternative becomes increasingly costly.

Empirical convergence. Systems that approximate A/M stratification—elastic weight consolidation [14], progressive neural networks, complementary learning systems [15]—outperform flat architectures on continual learning precisely because they protect encoded actions from learning interference. The trajectory from flat parameter spaces toward stratified architectures is the convergence this theorem predicts. □

Corollary 4 (Distinct Roles)**.** *Type irreducibility establishes distinct functional roles: A navigates W given K; M evolves (A, K) in response to learning signals. Concrete instances: neuroplasticity (M on A), insight (M on K), habit formation (M promoting from L_n to lower levels per Definition 12), evolution (M on M at the most stringent rigor threshold).*

6 Necessity of γ and η_M

Theorem 5 (Structural Invariants of Persistence)**.** *For any entity E persisting (Definition 11) under bounded active context C_n, survival threshold U_{lethal}, with positive novelty rate $\nu(E) > 0$:*

(i) $\gamma(E) > 0$ necessarily.

(ii) $\eta_M(E) > 0$ necessarily (in expectation over the entity's trajectory).

(iii) Every other scalar property of E except positive sensing capacity (a precondition of σ) can equal zero for some persistent entity.

(iv) Therefore $I(E) = \gamma \cdot \eta_M$ is the natural measure of intelligence: the rate of adaptation to novelty, identified by what positive $\nu(E)$ selects for.

When $\nu(E) = 0$, the entity faces no novel states and $\gamma = 0$ is viable: no free context is needed because nothing unexpected arrives. Such entities persist without intelligence. Intelligence is what positive novelty rate selects for.

Scope note. This theorem applies to entities with the structure $E = (K, A, \sigma, \gamma)$. This is a presupposition, not a conclusion: the theorem does not derive the *existence* of agency from non-agentive precursors. What it establishes is an *elimination result*: among entities that have this structure, those with $\gamma = 0$ do not persist under $\nu > 0$. The contribution is the *mechanism*—free-context exhaustion—not the origin of $\gamma > 0$ (which remains open; Section 10). Inert objects lack this structure entirely and are outside the theorem's domain.

Proof. **Part 1: Necessity of $\gamma > 0$: World Drift and Context Exhaustion.**

Suppose $\gamma(E) = 0$. By Definition 2, the entity never engages M: $(A(t), K(t)) = (A(0), K(0))$ for all t.

The entity has $\nu(E) > 0$: it encounters states where $U(w, K) > 0$ at positive rate. Under positive entropy, the world drifts: states that $K(0)$ was calibrated for become less frequent, and states not covered by $K(0)$ become more frequent. The entity must still act to survive. Each encounter with a novel state—one where no lower encoding level has an adequate response—requires real-time processing at L_n, consuming active context capacity.

An entity with $\gamma > 0$ resolves these encounters: it updates K, promotes proven actions to lower levels, and frees L_n for the next problem. Each resolution converts a novel state into a state handled below L_n. An entity with $\gamma = 0$ cannot resolve anything. The same types of novel states keep arriving and keep requiring L_n processing. As the world drifts further from $K(0)$, an increasing fraction of the entity's encounters are novel—$K(0)$ covers less of current W—so the concurrent demand on L_n grows. $C_{\text{free}}(t) \to 0$ not because K grows (it is fixed) but because the fraction of encounters requiring active context increases while no encounter is ever resolved.

When $C_{\text{free}} = 0$, the entity cannot load the relevant subgraph $K^{[a]}$ for any new problem: inference quality $\rho \to 0$. The entity encounters w where $U(w, K(0)) > U^d_{\text{lethal}}$ and does not survive.

The argument is self-contained at the individual level and requires no competition. For any initial $K(0)$, because $\nu > 0$ and the world drifts, there exists a finite time T^* at which the fraction of encounters requiring L_n exceeds what C_n can service. After T^*, the entity cannot process novel states at the rate they arrive. Since $\nu > 0$ continues, the entity eventually encounters a lethal state. No competitor is required: the environment alone is sufficient.

The critical distinction is between $\nu = 0$ and $\nu > 0$. An entity in a zero-novelty environment (a thermostat in a stable room) faces no context pressure and persists with $\gamma = 0$. The framework does not predict it dies. It predicts it does not need intelligence.

Positive entropy in W is the upstream cause of positive novelty rate: energy gradients produce states not contained in any finite K [5, 1]. But entropy is a property of the environment; novelty rate is a property of the entity–environment coupling. Entropy guarantees that $\nu > 0$ eventually for any entity whose niche is not permanently isolated from the rest of W.

Therefore $\gamma(E) > 0$ is necessary for persistence under $\nu > 0$.

Part 2: Necessity of $\eta_M > 0$.

Suppose $\gamma > 0$ but $\eta_M \leq 0$.

Case $\eta_M = 0$: The entity engages M but M produces no improvement in grounding. (A, K) is effectively fixed despite engagement. The free-context exhaustion argument applies: novel states consume context that is never freed by promotion. Additionally, C_n is bounded. Promotion to lower encoding levels requires achieving the rigor threshold θ_i (Definition 12), which requires M to improve the relevant action to the required certainty. With $\eta_M = 0$, no action reaches threshold, L_n is never relieved, and $\rho \to 0$.

Case $\eta_M < 0$ *persistently*: (A, K) worsens in expectation, accelerating the failure mode.

The persistence condition is:

$$\mathbb{E}_t[\eta_M] > 0 \quad \text{over the entity's trajectory}$$

not $\eta_M(t) > 0$ at every time step. Temporary $\eta_M < 0$ (exploration, creative destruction) is compatible with persistence.

Part 3: Minimality.

For every other scalar property of E, there exists a persistent entity for which that property equals zero:

- $K(0) = \varnothing$: built from monotone descent (Theorem 1).
- $A(0) = \{\text{id}\}$: the identity action still returns feedback from W.

- $C_n = C_{\min}$: graph structure [17] maintains $\rho = 1$ by retrieving the relevant subgraph.

The exception is sensing capacity: $\gamma > 0$ requires learning signals, which require that the selection mechanism σ be coupled to W via sensing. But sensing is excluded from the invariants not by definitional fiat but by structural role: it is a *precondition* for the entity to couple to W at all. The tuple (K, A, σ, γ) with zero sensing reduces to a static object that neither observes nor acts. The theorem's domain is entities that interact with W, which requires positive sensing as a precondition of interaction, not as a variable property that selection acts upon. The distinction is between preconditions (without which the framework does not apply) and invariants (properties that vary within the framework and whose values are selected for).

Part 4: $I(E) = \gamma \cdot \eta_M$ follows by arithmetic.

Parts 1–3 are the substantive content. Given that γ is a probability and η_M a conditional rate, their product is the expected improvement rate by the law of total expectation (Corollary 6). □

7 $I(E) = \gamma \cdot \eta_M$ Is Structurally Necessary

Corollary 6 (The Intelligence Measure)**.** *γ and η_M are the unique structural invariants of persistence (Theorem 5). The expected improvement rate is:*

$$I(E) = \gamma \cdot \eta_M$$

Proof. $\gamma \in [0, 1]$ is the marginal probability of engaging M. η_M is the expected improvement rate conditional on engagement. By the law of total expectation:

$$\mathbb{E}[\text{improvement rate}] = \gamma \cdot \mathbb{E}[\text{rate} \mid \text{engaged}] + (1 - \gamma) \cdot 0 = \gamma \cdot \eta_M$$

This identity holds whether γ is content-blind or directed. The conditioning in Definition 9 absorbs the selection effect. The product is exact.

The substantive claim is not the formula but the identification: intelligence is the rate at which an entity *adapts to novelty*, not the level of its current performance. The derivation makes this precise: $\gamma > 0$ and $\eta_M > 0$ are required specifically because $\nu(E) > 0$—novel states arrive that would be lethal without adaptation. Intelligence is what positive novelty rate selects for. This distinguishes the present account from Hutter [13] (intelligence as performance level) and from ordinal rankings that treat intelligence as a comparison without units. Given this identification, the product follows uniquely by total expectation: γ is a probability, η_M is a conditional rate, and their product is the unconditional rate. Any monotone transform $f(\gamma \cdot \eta_M)$ would measure a *function of* the rate, not the rate itself, and would lack the dimensional interpretation (bits of improvement per learning signal). The uniqueness is not that the product is the only function zero on the axes—many are—but that it is the only function that *is* the expected improvement rate rather than a rescaling of it. □

Temporal note. $I(E)$ is the expected improvement per learning signal, not per unit time. The temporal rate of improvement depends on the entity-environment coupling: if learning signals arrive at rate $\lambda(t)$, the instantaneous improvement rate is $I(E) \cdot \lambda(t)$. Two entities with identical $I(E)$ in different environments improve at different temporal rates. $I(E)$ measures the entity's intrinsic capacity; the environment determines how often that capacity is exercised.

Remark 3 (Exploration and Creative Destruction). Temporarily negative η_M—locally irrational steps that reduce grounding in the short run—is survival-optimal when the persistence condition requires only $\mathbb{E}_t[\eta_M] > 0$ over the trajectory. An entity restricted to $\eta_M \geq 0$ at every step is trapped in local optima; an entity that can tolerate temporary $\eta_M < 0$ accesses a larger region of the (A, K) landscape.

8 Engagement with Prior Work

8.1 Intelligence Measures and Definitions

Dupoux, LeCun, and Malik [6]. They deliberately avoid defining intelligence, instead characterizing it as the capacity for autonomous learning. Theorem 5 establishes that this is structurally correct: $\gamma > 0$ and $\eta_M > 0$ are the unique invariants. Their product is the definition they avoided giving. Their bilevel optimization framework (inner loop as development, outer loop as evolution) captures the correct structural relationship between encoding levels, but treats the learning rate difference as a design parameter: the inner loop is fast because it is designed to be fast. Definition 12 derives this from first principles: $\eta_{\text{evo}} \ll \eta_{\text{ind}}$ because $C_0 \gg C_n$. Their "System M" that is "not learned through gradient descent" but fixed by evolution is explained by $\theta_0 \approx 1$: only mechanisms validated across maximal environmental variation meet this threshold.

Hutter [13] and Legg–Hutter [16]. Three structural differences: (1) AIXI assumes unbounded computation; we have bounded C_n. Under bounded context, the AIXI solution is unreachable: the entity descends with finite resources (Theorem 1). (2) They define intelligence as performance level; we define it as rate of improvement, independent of current performance. (3) Their environment class is all computable functions; ours is physical W with positive entropy.

Chollet [4]. Skill-acquisition efficiency is structurally close to η_M. Two differences: Chollet treats prior knowledge K as a confound to be controlled for; here K is constitutive, because η_M operates on (A, K) and the efficiency of learning depends on what is already known. Chollet's "scope" and "generalization difficulty" map to the growth of reachable territory in the (A, K) landscape.

Schmidhuber [22]. Curiosity as compression progress corresponds to η_M. The difference is foundational. Schmidhuber treats intrinsic drive as a design objective. Theorem 5, Part 1, establishes $\gamma > 0$ as a necessary condition of persistence: not a reward to be designed in, but a property that selection enforces.

Cattell [3] and Spearman [24]. Cattell's fluid intelligence (G_f) maps to η_M; crystallized intelligence (G_c) maps to $|K|$. The framework predicts the well-established correlation: entities with higher η_M build larger K faster. Spearman's g-factor should correlate with $I(E) = \gamma \cdot \eta_M$—a testable prediction.

8.2 Self-Organization and Minimization

Friston [9, 10]. The free energy principle derives minimization behavior from self-organization: systems that persist must minimize surprise (free energy). This is consistent with monotone descent on $H(W \mid K)$ (Theorem 1). FEP does contain an elimination argument: systems that fail to minimize free energy cease to exist as identifiable systems. However, FEP does not distinguish between $\gamma = 0$ and $\gamma > 0$ within its framework: a rock trivially minimizes free energy, and a thermostat minimizes it with fixed (A, K). The present account adds a specific mechanism—free-context exhaustion under positive novelty rate—that eliminates $\gamma = 0$ entities. The contribution is

the context-filling argument (Theorem 5, Part 1): not that minimization occurs, but that active, learning-driven minimization ($\gamma > 0$) is necessary when $\nu > 0$.

8.3 Encoding Hierarchy in Practice

Hinton and Nowlan [11]. They demonstrate via simulation that individual learning can accelerate evolutionary search: the Baldwin effect. Our framework formalizes this: individual learning discovers actions at L_n; encoding permanence (Definition 12) determines which promote to L_0. The Baldwin effect is a necessary consequence of the encoding hierarchy.

Kirkpatrick et al. [14]. Elastic weight consolidation assigns higher importance to weights encoding well-established knowledge, penalizing changes during subsequent learning. This is the encoding hierarchy: weights at lower levels (higher θ_i) are protected; weights at higher levels are free to change. EWC's Fisher information matrix empirically estimates our C_i. The result provides direct evidence that artificial systems violating the hierarchy (flat learning rate) fail, and systems approximating it succeed.

Murray et al. [20]. Intrinsic timescales of neural activity increase along the cortical hierarchy: sensory cortex fluctuates rapidly, prefrontal cortex maintains stable representations. This maps onto the encoding hierarchy: fast-updating sensory areas at L_n, slow-updating prefrontal areas at lower levels. The empirical timescale gradient matches $\eta_i \propto 1/\log(C_i/\varepsilon)$.

Finn et al. [8] and Thrun [25]. MAML and "learning to learn" are M operating on M: the $M^{(2)}$ level. The meta-learner updates more slowly ($\eta_{M^{(2)}} < \eta_M$) because $C_{M^{(2)}} > C_M$. The MAML inner loop is L_n; the outer loop is L_{n-1}. Evolution is meta-meta-...-learning at L_0.

9 The Complete Chain

The four papers derive the necessary architecture of intelligence from a single generative constraint:

1. **Bounded C_n + information preservation $\Rightarrow$ graph structure** [17]. Any entity accumulating propositions under bounded context necessarily maintains a directed graph with typed edges.

2. **Bounded C_n + growing A $\Rightarrow$ convergence to escalation** [18]. Any viable control architecture converges toward escalation properties as concurrent demands grow relative to C_n.

3. **Bounded context $\Rightarrow$ normalization of shared propositions into K, with asymmetric retention** [19]. Overlapping action policies under bounded C_n force extraction of shared propositions into separated K; the retention criterion is probability of need $\times$ conditional value.

4. **$\gamma > 0$ + feedback $\Rightarrow$ descent on U + encoding hierarchy; positive entropy $\Rightarrow$ $\gamma > 0$ and $\eta_M > 0$ necessary; $I(E) = \gamma \cdot \eta_M$** [this paper]. Agency with feedback produces convergent uncertainty reduction. Cost-of-failure generates the encoding hierarchy. The lineage and context-filling arguments establish the two invariants; their product is the unique intelligence measure.

Each step is a necessary consequence of bounded context interacting with properties of the physical world: positive entropy, survival pressure, and feedback.

10 Open Questions

1. **Origin of γ.** Theorem 5 establishes that $\gamma > 0$ is necessary but does not explain how $\gamma > 0$ structures bootstrap from $\gamma = 0$ substrates. The thermodynamic conjecture: dissipative structures [21, 7] under sustained energy flow spontaneously produce $\gamma > 0$ configurations, which selection retains.

2. **Direction of γ.** Is agency uniformly distributed across learning signals, or does it have intrinsic directedness?

3. **The $M^{(n)}$ hierarchy.** How many levels of meta-learning does a persistent entity require? Is there a natural termination point, or is the hierarchy open upward without bound?

4. **Bound composition in A.** How do individual action bounds compose when multiple actions share knowledge substructure?

11 Falsifiable Predictions

The theorems yield eight predictions that are empirically testable.

1. Catastrophic forgetting will not be solved within flat parameter spaces. Theorem 3 predicts that any continual learning solution that works at scale will implement A/M stratification—whether explicitly (elastic weight consolidation, progressive networks, complementary learning systems) or implicitly (by learning to protect certain parameters from learning updates). A solution that uses a single undifferentiated parameter space for both action execution and learning, with no form of weight protection or functional separation, will fail on long-horizon continual learning benchmarks (100+ sequential tasks). A flat-parameter system that matches stratified architectures on such benchmarks without any form of functional separation would falsify Theorem 3.

2. Effective agent architectures will converge to escalation. Multi-step AI agents (planning-based systems that decompose goals into subtask sequences) operate under bounded C_n: each delegation requires tracking context. Theorem 5 and the encoding hierarchy (Definition 12) predict that successful agent architectures will converge toward escalation: routine subtasks handled by compiled, cached, or tool-based policies at lower encoding levels, with the frontier model invoked only on failure or novelty. Purely top-down planning agents that maintain full supervisory state over every subtask will fail to scale beyond $k^* = C_n/(2\alpha)$ concurrent delegations. A purely top-down planning agent that scales to 1000+ step tasks without delegating to compiled subroutines would falsify the escalation convergence result.

3. $I(E) = \gamma \cdot \eta_M$ predicts G_f correlations. If intelligence is the rate of improvement, then Cattell's fluid intelligence G_f (which measures novel problem-solving capacity) should correlate more strongly with η_M than with $|K|$. The prediction: across individuals, G_f scores will predict learning rate on novel tasks better than they predict current knowledge volume. Crystallized intelligence G_c should correlate with $|K|$ and with the cumulative integral of $I(E)$ over time. A population in which G_c predicts novel-task learning rate better than G_f does would falsify the identification.

4. $\gamma = 0$ systems fail under distribution shift. The free-context exhaustion argument (Theorem 5, Part 1) predicts that systems with $\gamma = 0$ (no capacity to update K or A in response to feedback) will fail under sustained distribution shift, even if they perform well on the initial

distribution. The mechanism is specific: $C_{\text{free}} \to 0$ as unresolvable novel states accumulate. The prediction: deployed ML systems without online learning capabilities will show monotonically degrading performance under distribution drift, with the degradation rate proportional to the novelty rate ν. A static model that maintains performance indefinitely under sustained distribution shift would falsify the free-context exhaustion argument.

5. Graph-structured knowledge will produce cross-specialty unifications. The graph necessity result of [17] establishes that shared propositions are factored into reusable nodes with directed typed edges. When knowledge from different specialties is encoded in a shared graph, propositions that were independently maintained in separate domains become structurally adjacent—their shared dependencies become visible. The prediction: as institutional and scientific knowledge graphs grow, cross-specialty unifications (where two fields discover they are studying the same underlying structure under different names) will increase in frequency, driven not by individual insight but by the graph making shared structure visible. Falsified if large-scale knowledge graphs produce no increase in detected cross-domain structural equivalences compared to unstructured literature search.

6. Open knowledge graphs with provenance will undercut parametric knowledge storage. The normalization result [19] establishes that commonly held knowledge—the high-overlap region of K across entities—is exactly the set of propositions that factoring extracts into shared, reusable nodes. Provenance is a typed edge: each proposition carries its reconstruction path to source evidence [17]. An open-source knowledge graph encoding commonly held propositions with provenance would provide $O(d)$ retrieval with full attribution at near-zero marginal cost. Current large language models store the same commonly held knowledge in continuous weights: retrieval is $O(n^2)$, provenance is absent, and the knowledge is neither inspectable nor correctable. The prediction: open knowledge graphs with provenance will replicate and eventually surpass the factual-knowledge function of parametric models, at a fraction of the compute cost, because structured retrieval with attribution is strictly cheaper than re-deriving answers from weights on every query. Falsified if parametric storage achieves cost parity with structured retrieval on factual knowledge tasks at arbitrary scale while simultaneously providing equivalent provenance.

7. The first widely recognized artificial general intelligence will have explicit graph structure, not monolithic parameters. Theorem 5 identifies the structural invariants of intelligence as $\gamma > 0$ (agency) and $\eta_M > 0$ (learning efficiency). Current large language models have $\gamma = 0$: they do not update K or A in response to feedback during deployment. They approximate intelligence by storing a large static K in weights and simulating $\gamma > 0$ within the context window—a strategy the free-context exhaustion argument (Theorem 5, Part 1) predicts cannot persist under positive novelty rate. The framework predicts that the first system widely recognized as artificial general intelligence will exhibit: (i) explicit or learned graph structure over propositions [17], (ii) escalation-based control [18] rather than monolithic inference, (iii) separated K with provenance rather than knowledge fused into parameters, and (iv) $\gamma > 0$—genuine online learning that updates (A, K) from feedback. These are not design recommendations but convergence results: any system that persists under bounded context and positive novelty rate will exhibit these properties regardless of substrate. Falsified if a system with monolithic parameters, no graph structure, no knowledge separation, and no online learning achieves robust, open-ended performance across novel domains over extended deployment without degradation.

8. Cancer treatment will converge on restoring broken escalation chains. The escalation hierarchy [18] predicts that multicellular organisms handle errors at the lowest sufficient level: DNA repair (L_0), cell-cycle arrest (L_1), apoptosis (L_2), immune surveillance (L_3), with conscious medical intervention at L_n. Cancer is the pathology of broken escalation: mutations disable one

or more lower-level error handlers, so cellular errors that should be caught and resolved locally instead escalate unchecked to the organism level. Current treatment paradigms attack the cancer directly at L_n (surgery, chemotherapy, radiation)—the equivalent of top-down control. The escalation framework predicts that effective treatment will converge toward restoring the broken levels of the chain rather than compensating at L_n. Checkpoint immunotherapy (restoring immune surveillance), synthetic lethality approaches (exploiting remaining repair pathways), and p53 reactivation therapies are early instances: each restores a lower-level handler rather than attacking the tumor directly. The prediction: treatment efficacy will correlate with the number of escalation levels restored, not with the directness of tumor targeting, and the field will shift from "kill the cancer" toward "repair the chain that should have caught it." Falsified if direct tumor-targeting therapies consistently outperform escalation-restoring therapies across cancer types in long-term survival outcomes.

12 Conclusion

Intelligence is the rate of adaptation to novelty. This is not proposed but identified by elimination: positive novelty rate requires $\gamma > 0$ and $\eta_M > 0$; their product is the unconditional rate.

Theorem 1 establishes that agency with feedback produces monotone local descent on expected uncertainty $H(W \mid K)$. Each informative update reduces expected uncertainty by at most the information gain, with equality under exact Bayesian conditioning. Under non-stationary task distribution, descent is local and instantaneous rather than globally convergent.

Definition 12 derives the encoding hierarchy from cost-of-failure: $\theta_i = 1 - \varepsilon/C_i$ determines the certainty required for promotion, and $\eta_{\text{evo}} \ll \eta_{\text{ind}}$ follows from $C_0 \gg C_n$.

Theorem 3 establishes that A/M stratification is selectively favored under bounded context: collapsing action execution and learning into a single parameter space is possible but increasingly costly under positive novelty rate.

Theorem 5 establishes that $\gamma > 0$ and $\eta_M > 0$ are the unique structural invariants of persistence under positive novelty rate. The free-context exhaustion argument eliminates $\gamma = 0$ entities; the promotion argument eliminates $\eta_M = 0$ entities. The product $I(E) = \gamma \cdot \eta_M$ is the unique measure consistent with these invariants, derived from the law of total expectation applied to a probability gate and a conditional rate.

The four-paper chain derives, from the single constraint of bounded active context, the necessary architecture: graph structure, escalation, knowledge–action inseparability, evaluation-driven descent with encoding permanence, and the structural invariants of intelligence. Each result is a necessary consequence of bounded context interacting with properties of the physical world: positive entropy, survival pressure, and feedback.

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